

# Classmoji: A GitHub-Native LMS for Flexible CS Education

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## Abstract

*Classmoji* is a learning management system (LMS) built on GitHub's infrastructure for computer science courses. Traditional LMS platforms are designed around file-based submissions, but this model fails to leverage industry-standard development workflows that rely on version control, issue tracking, and collaborative repositories. As a result, instructors often end up managing multiple systems or asking students to submit their work in ways that differ from real-world software development practices.

*Classmoji* addresses this mismatch by treating GitHub organizations as classrooms, repositories as course modules, and issues as assignments. This direct mapping enables seamless course management within a platform already familiar to students and widely used in industry, introducing three features tailored to educational contexts: (1) automated creation and management of repositories and issues; (2) an emoji-based feedback system that replaces numerical scores; and (3) a token-based time management mechanism that gives students flexible control over assignment deadlines.

## Keywords

github, alternative grading, computer science education, learning management system, student assessment

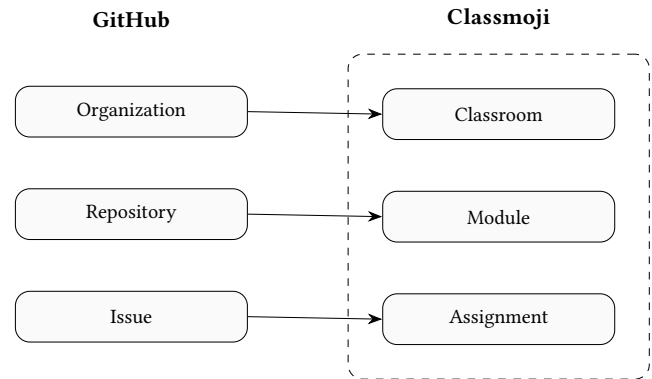
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## 1 Introduction

Traditional learning management systems (LMS) such as Canvas, Blackboard, and Moodle are designed for document-based coursework, where students typically submit PDFs or other static file formats. In computer science courses, however, instructors often ask students to compress their code into ZIP files for submission, a practice that contradicts standard software development workflows and fails to leverage version control and collaboration that are central to professional programming.

Recognizing this mismatch, many CS educators have adopted Git and GitHub for coursework [2, 4, 5], using repositories to manage projects, submissions, and team collaboration. While this aligns



**Figure 1:** *Classmoji* maps GitHub concepts to LMS concepts and adds course features (roster sync, deadlines, grading) on top of GitHub APIs to create a GitHub-native LMS.

coursework with real-world software development, it creates practical challenges: instructors must manage many repositories, coordinate grading across teaching assistants, and handle deadlines in platforms not designed for course management. Moreover, growing interest in alternative grading approaches [1, 3] has exposed further limitations of traditional LMSs, where rigid gradebooks force workarounds for qualitative or outcomes-based assessment.

These gaps motivated the development of *Classmoji*<sup>1</sup>, a GitHub-native LMS that integrates real-world software development workflows with flexible grading approaches.

## 2 Related Work

Several tools have sought to support GitHub-based coursework, including ACAD<sup>2</sup>, Gitomator<sup>3</sup>, GHClass<sup>4</sup>, Rhomboid<sup>5</sup>, Virtual Classroom<sup>6</sup>, and Repobee<sup>7</sup>. These projects demonstrate the viability of GitHub as an educational platform, but most are command-line only, which limits their accessibility and ease of use for course management. Many also suffer from maintenance challenges, with some projects showing little or no activity in recent years. GitHub Classroom<sup>8</sup> is the most widely adopted alternative, offering an instructor-friendly interface and active maintenance. However, it functions as a supplemental tool rather than a comprehensive learning management system, requiring parallel workflows across multiple platforms.

<sup>1</sup><https://classmoji.io>

<sup>2</sup><https://github.com/UCSB-CS-Using-GitHub-In-Courses>

<sup>3</sup><https://github.com/gitomator>

<sup>4</sup><https://github.com/rundel/ghclass>

<sup>5</sup><https://github.com/mgelbart/rhomboid>

<sup>6</sup><https://github.com/hplgit/virtual-classroom>

<sup>7</sup><https://github.com/repobee>

<sup>8</sup><https://classroom.github.com/>



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Beyond GitHub infrastructure challenges, the rise of alternative grading practices [1, 3] has exposed further limitations of traditional LMSs. Conventional gradebooks are built around numerical scores and letter grades, making it difficult to implement qualitative feedback systems that emphasize learning outcomes over point accumulation. Few platforms provide native support for non-numerical assessment approaches.

*Classmoji* builds on these prior efforts while addressing both their infrastructure and grading limitations, providing a complete GitHub-native LMS with automated repository management and support for flexible assessment practices.

### 3 Classmoji

*Classmoji* is a web-based learning management platform designed for computer science courses. The system leverages GitHub’s organizational structure and development workflows to create an educational environment that mirrors real-world software development practices. The platform addresses several challenges observed in our courses: the complexity of managing GitHub-based coursework at scale, the mismatch between granular numerical grading and project-based assessment, and the inflexibility of traditional late policies. *Classmoji* supports three primary roles: instructors, students, and teaching assistants. Each role has a tailored interface to support its responsibilities.

#### 3.1 GitHub-Native Course Infrastructure

The platform automates the creation and management of GitHub repositories and issues. When instructors publish a module, the system generates individual student repositories from templates with starter code or shared repositories for team assignments. The platform links academic identities to GitHub accounts: instructors upload a CSV roster, and students connect their GitHub accounts using school credentials.

Assignments are represented as GitHub issues, automatically created when modules are published. Each assignment includes release dates, submission deadlines, and grader deadlines. Submission timestamps are determined when students close their issues, mirroring the professional practice of closing issues to mark task completion.

#### 3.2 Emoji-Based Grading

Recognizing that project-based CS courses do not always require fine-grained numerical distinctions, *Classmoji* implements emoji-based grading to shift student focus away from scores and toward qualitative feedback. Instructors can select any set of emojis and map them to underlying numerical grades that remain hidden from students. Each emoji can include descriptive feedback explaining what the performance level represents. Assignments may be distributed randomly among graders to balance workload, or consistently assigned so that the same grader evaluates all work for a given student.

#### 3.3 Token-Based Time Management

Traditional late-day policies are replaced with a flexible token economy. Instead of requiring students to spend entire 24-hour extensions when they only need a few hours, instructors provide a pool of

tokens at the start of the term that can be redeemed for hour-based deadline extensions. Instructors set the token cost per hour for each assignment, giving students precise control over the amount of extra time they use. Students can allocate tokens strategically, request refunds to reallocate them to other assignments, and earn extra tokens by completing optional assignments, encouraging engagement and flexibility.

#### 3.4 AI-Powered Assessment

The platform integrates AI-driven assessment tools that extend beyond traditional auto-grading. Instructors specify learning objectives and topics, then select between two quiz modes: standard knowledge checks or code-aware assessments that explore student repositories before formulating questions. The code-aware mode clones student code, analyzes implementations, and generates contextual questions probing understanding of design decisions and implementation choices. The system provides structured feedback with credit-based scoring and tracks student engagement metrics during assessment sessions.

#### 3.5 Additional Features

The platform also provides tools for managing teaching assistants, tracking grading progress, and distributing course materials. Students can resubmit work based on feedback, and instructors can share syllabi and lecture slides directly within the system.

### 4 Future Work

*Classmoji* has been under active development since 2023 and deployed across four semesters at two institutions. Our research agenda focuses on user studies examining student and instructor experiences through surveys and interviews on the emoji-based grading system, token economy, and workflow integration, guiding a longer-term study of learning outcomes and engagement patterns. A key priority is evaluating our AI-powered assessment features in classroom settings to understand their impact on student learning and code comprehension compared to traditional assessment methods. By presenting this poster, we aim to gather feedback from the community to refine these directions and ensure that *Classmoji* continues to evolve in ways that address instructors’ classroom needs.

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